Simple and Easy Questions

a) Write a C++ program that defines a structure to store a student's name, 5 test scores, and calculates the average score. The program should prompt the user for the student's name and test scores, then display the name, the scores (with two decimal places), and the average score (also formatted to two decimal places).

Example:

Input:

```mathematica
Enter the name of the student: John Doe
Enter the 5 test scores: 85 90 78 92 88
```

Output:

```yaml
Student name is: John Doe
Test scores: 85.00 90.00 78.00 92.00 88.00
The average test score is: 86.60
```

b) Write a C++ program to simulate a restaurant ordering system where the user can select from 8 breakfast items, specifying the quantity for each. The program should display a receipt with the quantity, item name, price, total cost, tax (5%), and the final amount (including tax). The user can continue selecting items or exit. Proper formatting should be applied for the amounts.

Input:

```mathematica
Enter your choice: 1
Enter the Quantity: 2
Enter (y/n) to select more items: y
Enter your choice: 5
Enter the Quantity: 1
Enter (y/n) to select more items: n
```

Output:

```markdown
********RECEIPT********

2   Plain Egg 1.45$
1   Fruit Basket 2.49$
    Tax      0.22$
***********************************************
Total Amount is  4.16$
***********************************************
```

c) Write a C++ program to calculate the billing price for Residential and Business customers.

- **Residential bill includes a $4.50 processing fee, $20.50 basic service fee, and $7.50 per premium channel.**
- **Business bill includes a $15.00 processing fee, $75.00 basic service fee for up to 10 connections (plus $5 for each additional connection), and $50.00 per premium channel.**

The program should prompt for customer type, account number, premium channels, and connections (if applicable), then display the total bill.

Input:

```mathematica
Enter your choice: r
Enter the customer's account number: 123
Enter the number of premium channels: 3
```

Output:

```swift
The customer account no is: 123
Total Billing Price of Residential customer is: $49.00
```

d) Create a program simulating a glass with an initial liquid level of 200 ml. The user can request a specific amount of drink. If the remaining liquid is less than 100 ml, the glass is refilled automatically to 200 ml. After refilling, the user is prompted whether they want to take another drink or not. The process repeats until the user chooses to stop.

Example:

Input:

```vbnet
Enter the level of drink in ml the user required: 150
Enter the level of drink in ml the user required: 50
Do you want to take drink again? Press 1 for it else it terminate
```

Output:

```vbnet
The 50 ml of drink is left now.
The drink is refilled to 200 ml.
```

e) Design a program to manage book information. The program should store the details of 3 books including the book's name, ISBN, author's name, and publisher's name. After entering the book details, the user can search for a book by its ISBN. If a matching ISBN is found, the program will display the book's details.

Input:

```yaml
Enter name of book: C++ Programming
Enter ISBN: 1234
Enter author name: John Doe
Enter publisher name: ABC Publishers

Enter ISBN number to search a book: 1234
```

Output:

```yaml
Name: C++ Programming
ISBN: 1234
Author name: John Doe
Publisher name: ABC Publishers
```

f) Write a C++ program to calculate the area and perimeter of a circle. The program should:

1. Use a parameterized constructor to initialize the radius.
2. Calculate and display the area ($\pi r^{2)}$ and perimeter ($2\pi r$) with two decimal places.
3. Demonstrate the use of a destructor.

Input:

```mathematica
Enter the radius: 5
```

Output:

```csharp
The area is: 78.55
The perimeter is: 31.42
```

g) Create a C++ program to simulate a basic banking system with the following features:

1. Add Balance: Deposit a specified amount into the account.
2. Withdraw Balance: Withdraw a specified amount, checking if the balance is sufficient.
3. Show Balance: Display the current balance.
4. The program should repeatedly display a menu for the user to choose an action until an invalid option is entered.

The initial account balance is 9999.

Input:

```mathematica
1 (Add Balance)
Enter the amount of money you want to add: 500
2 (Withdraw Balance)
Enter the amount of money you want to withdraw: 300
3 (Show Balance)
```

Output:

```csharp
The current balance after adding money is: 10499
The Balance after Withdrawal of money 300 is: 10199
The current amount left in your account is: 10199
```

h) Write a C++ program that simulates generating an invoice for a hardware store. The program should:

1. Ask the user for the number of products.
2. For each product, input the name, description, quantity, and price.
3. Display each product's details (name, description, quantity, and price).
4. Calculate and display the total amount for all products, with the price displayed to two decimal places.

The program should include a class invoice that handles input and calculates the total. Here's an example output:

i) Write a C++ program to manage a book inventory. The program should:

- Display Books: Show a list of books with their author, title, price, publisher, and stock.
- Search and Buy Books: Allow the user to search by author and title. If available, let the user specify how many copies to buy. If the quantity exceeds stock, prompt for a valid amount and show the total cost.
- Menu Options: Provide options to display books, search, or exit.

Example output:

```mathematica
MENU
1. Display books
2. Search book
3. Exit

Enter choice: 1
Book author        Book title        Price  Publisher  Stock
Jkrolling          Harry Potter      3000   xyz        5

Enter choice: 2
Enter author: Jkrolling
Enter title: Harry Potter
The book is available with 5 copies in stock.
Enter quantity: 3
Total: 9000
Do you want to buy another? (1 for yes): 1
```

j) Create a Coffee Shop Order Management System in C++. The system should allow customers to view a menu, place orders, calculate totals, and fulfill orders.

Requirements:

1. **Menu Display:** Show three items (Pizza, Pasta, Coffee) with their prices.
2. **Order Placement:** Customers select an item and enter the quantity. The system calculates the total cost.
3. **Order Fulfillment:** Track if orders are fulfilled. Display the order details, including item name, quantity, and total amount.
4. **Output:** Confirm order readiness or show a message if all orders are fulfilled.

Example:

```mathematica
Menu:
1-pizza  food   200
2-pasta  food   300
3-coffee drink  65

Enter item number: 1
Enter quantity: 2
Order: Pizza, Quantity: 2, Amount: 400
Order is ready.
```

k) Create a C++ program to input scores for 5 students, calculate their total score, and display it.

Requirements:

1. Input the scores for 5 students.
2. Calculate the total score.
3. Display the total score.

Example:

```mathematica
Enter the score of student 1: 85
Enter the score of student 2: 90
Enter the score of student 3: 78
Enter the score of student 4: 88
Enter the score of student 5: 92

Total Score = 433
```

l) Create a C++ program that defines a Circle class with a static variable to count the number of objects created. Each object should calculate and display the area of the circle based on its radius.

Requirements:

Class `Circle`**:**

1. Private members: `radius` (int).
2. Static member `count` to track the number of objects created.
3. Constructor that takes an integer `r` (radius), increments `count`, and displays the object number and the area.
4. `getarea()` method to calculate and return the area using πr2πr2 (with π≈3.142π≈3.142).

Main Function:

1. Input three radii from the user.
2. Create three `Circle` objects and display the object number and area for each.

Example Output:

```csharp
Enter the radius: 5
Enter the radius: 10
Enter the radius: 7

The object no that is created: 1
The area is: 78.55

The object no that is created: 2
The area is: 314.2

The object no that is created: 3
The area is: 153.94
```

m) Write a C++ program that demonstrates the use of a constant member variable and const member function.

Requirements:

Define a class `Demos` with:

1. A **constant integer** `x` initialized in the constructor.
2. A **const member function** `display()` that prints `"Inside the display function."` followed by the value of `x`.
3. A regular function `showMessage()` that displays `"Inside show message function."`.

In `main()`, create an object `d1`, initialize it with a value for `x`, and call `showMessage()` and `display()`.

n) Write a C++ program to demonstrate the use of static variables and static member functions.

Requirements:

Define a class `Add` with:

1. Two **static integer variables** `a` and `b`.
2. A **static member function** `sum()` that:
    1. Prompts the user to enter values for `a` and `b`.
    2. Calculates the sum of `a` and `b`.
    3. Displays the sum.

In `main()`, create an object of class `Add` and call the `sum()` function.

Sample Output:

```less
Enter the value of a: 10
Enter the value of b: 20
The sum is 30
```

o) Design a program that manages information about two books using inheritance. Create a base class Books with attributes for the book name, publisher name, ISBN, and genre. Then, create two derived classes B1 and B2 to input details for two books.

- B1 asks for details of the first book.
- B2 asks for details of the second book.
- In the main() function, display the details for both books

```yaml
Enter the genre of the book: Fiction
Enter the Book name: The Great Gatsby
Enter the publisher name: Scribner
Enter the isbn no: 9780743273565

Enter the genre of the book: Mystery
Enter the Book name: The Girl with the Dragon Tattoo
Enter the publisher name: Vintage
Enter the isbn no: 9780307949486

The genre of the Book 1 is: Fiction
The name of the Book is: The Great Gatsby
The publisher name is: Scribner
The isbn no of the book is: 9780743273565

The genre of the Book 2 is: Mystery
The name of the Book is: The Girl with the Dragon Tattoo
The publisher name is: Vintage
The isbn no of the book is: 9780307949486
```

p) Design a vehicle management system that models two types of vehicles: a car (Audi) and a motorcycle (Yamaha).

Classes:

1.) Vehicle:

Holds an integer array `price[2]` for the price of two vehicles: one car and one motorcycle.

2.) Car (Audi):

- Inherits from `Vehicle`.
- Attributes: `seatcapacity`, `nodoor`, `ftype`, and `mtype` (model type).

3.) Motorcycle (Yamaha):

- Inherits from `Vehicle`.
- Attributes: `nocy` (cylinders), `nog` (gears), `now` (wheels), and `maketype`.

Input Details:

- For Audi: Price, seat capacity, number of doors, fuel type, and model type.
- For Yamaha: Price, number of cylinders, gears, wheels, and make type.

Example Output:

```yaml
Enter the Price of the Audi vehicle: 50000
Enter the seating capacity of the car: 5
Enter the no of doors of the vehicle: 4
Enter the fuel type: petrol
Enter the model type: Q7

Enter the Price of the Yamaha vehicle: 15000
Enter the no of cylinders: 2
Enter the no of gears: 6
Enter the no of wheels: 2
Enter the make type: Sport

Details of Audi:
Price: 50000, Seats: 5, Doors: 4, Fuel: petrol, Model: Q7

Details of Yamaha:
Price: 15000, Cylinders: 2, Gears: 6, Wheels: 2, Make: Sport
```

q) Create a system to manage student information and calculate their total and average marks using inheritance.

Classes:

1.) Student:

Attributes: `name`, `cls`, `rollno`.
Constructor to input student details.

2.) Marks (Inherits from `Student`):

Attributes: `markse`, `marksm`, `marksp` (marks for English, Math, and Physics).
Constructor to input marks.

3.) Result (Inherits from `Marks`):

4.) Methods:
`infostd()`: Displays student info.
`tmarks()`: Returns total marks.
`avgmarks()`: Retrns average marks.

Input:

```mathematica
Enter the name: John
Enter the class: 10
Enter the roll no: 101
Enter marks of eng: 75
Enter marks of maths: 80
Enter marks of physics: 70
```

Output:

```css
John
10
101
The total marks are 225
The avg marks are 75
```

r) Create a system to manage student information and calculate their total and average marks using inheritance.

Long Questions

1.) How would you design the ISBN class to validate and parse the four parts of an ISBN string (e.g., 0 941831 39 6)? Implement the constructor and setISBN method to ensure the ISBN string is in the correct format.
How would you implement the Book class to store and display details such as title, author, publisher, price, and the associated ISBN? Write the setBookISBN and printDetails methods to interact with the ISBN class.
2.) Create a class named **EmployeeBonus**. The fields include the employee's salary, the bonus rate, and the total bonus. Create two constructors. Each constructor requires the salary (expressed as a double) and the bonus rate. One constructor requires the bonus rate to be a double, such as `0.1`. The other requires the salary and the bonus rate expressed as a whole number, such as `10`. Each constructor calculates the total bonus value based on the employee's salary multiplied by the bonus rate. The difference is that the constructor that accepts the whole number must convert it to a percentage by dividing it by 100.
Also, include a **displayBonusDetails** function that displays the salary, bonus rate, and total bonus for the employee.

Write a **main()** function that instantiates at least two **EmployeeBonus** objects—one that uses a decimal and one that uses a whole number as the bonus rate. Display the **EmployeeBonus** object values. Save the file as `EmployeeBonus.cpp`.

3.) Create a class named CarRental:

Include fields for a customer's data as an aggregate object. Create a Customer class that includes fields for name, license number, and contact number. Include a default constructor that initializes name, license number, and contact number to "Unknown" if no arguments are supplied. Also include a **displayDetails** function to display the customer's information. Write the class definition for a RentalDate class that contains three integer data members: pickupDay, pickupMonth, and pickupYear. Include a default constructor that assigns the date to 1/1/2023 for any new object that does not receive arguments. Also include a displayDate function to display the rental date.

Write a program that creates and displays:

A Customer object using the default constructor.

A RentalDate object using the default constructor.

Another set of Customer and RentalDate objects using parameterized constructors.